Portfolio summary

1. Which dataset, and which question does your project answer?

We used the Pistachio Image Dataset from Kaggle, containing JPG images of two pistachio types, Siirt and Kirmizi. All images have a resolution of 600 × 600 pixels, and the dataset was downloaded on 24 August 2026.

We initially selected a subset of pistachio images with similar characteristics for our experiments in P1. This provided a relatively consistent basis for investigating image properties, frequency distributions and the effects of filtering and downsampling.

In P2, we expanded our selection by introducing additional images with more varied characteristics. This was important because different image content can produce different frequency distributions, which can influence compression efficiency. By testing more varied images, we could investigate how well our compression algorithm performed across different types of image data, rather than only on similar pistachio images.

Our project focuses on how image data can be analyzed, processed, compressed and validated. In P1, we investigated image properties, frequency analysis, Gaussian filtering and downsampling. In P2, we developed and tested a lossless Huffman-based compression algorithm and compared its performance against other image formats. In P3, we developed a reproducible pipeline with validation checks, file hashes and a manifest to ensure data integrity.

The main question our project explores is how image data can be processed and compressed efficiently while preserving its integrity and ensuring reproducibility.

2. How did each part shape the next?

In P1, we examined the image properties and experimented with Gaussian filtering and downsampling. We found that downsampling reduced image detail, which highlighted the importance of preserving information when processing images.

In P2, we developed a lossless Huffman-based compression algorithm with different prediction methods. We used the image representation established in P1 and tested whether the images could be compressed and reconstructed without losing data. We also compared compression performance across different images. Since our algorithm uses lossless compression, we chose PNG as a baseline because it is a widely used lossless image format. This allowed us to compare our algorithm's compression performance against an established format while ensuring that the original image data was preserved.

In P3, we built on the compression algorithm from P2 by implementing it in a reproducible data pipeline. We added validation checks, SHA-256 hashes and a manifest to ensure that the images were processed correctly and that their integrity could be verified.

3. What is your strongest evidence, and the main limitation across the three parts?

Our strongest evidence is the successful lossless reconstruction of images in P2 and the validation results from P3. In P2, we tested our compression algorithm on both pistachio images and additional images with different characteristics, confirming that the original pixel data could be recovered without loss.

In P3, we successfully processed 10 images with no corrupt or invalid files. Our round-trip tests and manifest verification also passed, demonstrating that our pipeline can process and verify the tested images reliably.

And important finding from P2 was the compression depended on the characteristics of the images. Images with more noise and less predictable pixel patters were generally harder to compress, as they contained less redundancy for out prediction methods and Huffman coding to exploit. In these cases, the resulting compression become larger than the original images.

We also experimented with a spiral delta predictor. Since the images of the pistachio had a rough circular shape. However, it was not effective on the pistachio images because our automatic predictor selection continued to choose the vertical predictor instead. On images containing a complete circular pattern, the spiral predictor performed better. This demonstrated that different prediction methods are suited to different image structures, and that a more complex predictor does not necessarily improve compression for every type of image.

One limitation is that we have only tested a small selection of images. Although we included additional images in P2, we cannot be certain that our results represent larger and more varied datasets. Broader testing was outside the scope

of the project, and the limited variation between the available images made it difficult to evaluate performance across different image types.

4. Is the dataset fit for use: use, use with controls, investigate, or do not use? What would change it?

The Pistachio Image Dataset is suitable for our project because it provides images in a consistent format and resolution, allowing us to investigate image properties, prediction methods and lossless compression.

However, the dataset has some limitations. The images have limited variation in background, lighting and orientation, and there is little information about the camera settings or image capture conditions. This means that our results may not represent how the algorithms perform on images from more varied sources.

We therefore consider the dataset suitable for controlled experiments, but not sufficient to establish how well our methods would perform on image data in general. To reduce this limitation, we introduced additional images with more varied characteristics during P2 and validated the files used in P3.

A larger and more diverse dataset, with documented capture conditions and both pistachio classes represented consistently, would make the dataset more suitable for broader evaluation.

5. What is still unresolved?

There are still questions that our experiments do not fully answer.

Firstly, P3 tested only a small selection of Kirmizi pistachio images. More images from both the Siirt and Kirmizi classes would be needed to determine whether the processing pipeline works reliably across the entire dataset.

Secondly, although we experimented with different prediction methods in P2, we have not established how well they generalise to other types of images. For example, our spiral delta predictor performed better on an image with a more complete circular pattern, but the automatic predictor selection continued to favour the vertical predictor for the pistachio images, despite their roughly circular appearance. Further testing would be needed to understand which image characteristics make spiral prediction effective.

Finally, we have not tested the pipeline on a sufficiently large and diverse dataset to establish its reliability beyond our selected images. These questions would need to be investigated before drawing broader conclusions about the performance of our methods.

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